WEATHER APP

Project Understanding & Revision Guide

A beginner-friendly explanation of how the React application works, fetches weather data, and can be explained in a viva or interview.

Project type: Frontend web application
Core technologies: React.js, JavaScript, Axios, OpenWeather API, HTML and CSS.

1. Project overview

The Weather App lets a user enter a city and displays current weather information returned by OpenWeather. The browser sends a request over HTTP; the weather data comes from the external API, not from a local database.

Features

• Search for a city by typing its name and pressing Enter.
• Display city/country, temperature, weather description/icon, wind speed, and local date.
• Handle loading and failed requests.

2. End-to-end workflow

Step	What happens
1. User input	The user enters a city, such as Ahmedabad.
2. React state	useState stores the input and weather/loading/error state.
3. API request	Axios sends a GET request to OpenWeather with the city, units and API key.
4. JSON response	The API returns structured weather data, usually as JSON.
5. Render UI	The response is saved in React state; React refreshes the relevant interface.

3. Key code concepts

A. React state with useState

```
const [input, setInput] = useState('');\n\nconst [weather, setWeather] = useState({\n loading: false,\n data: {},\n error: false,\n});
```

input stores the city typed by the user. setInput updates it. weather.data holds the API response; loading indicates a request is in progress; error indicates a failed request. Updating state lets React refresh the interface.

B. Handling the input

```
onChange={(event) => setInput(event.target.value)}
```

This updates the stored city as the user types. The search handler checks for the Enter key before making the request, so it does not request weather after every keystroke.

C. Requesting weather with Axios

```
const url = 'https://api.openweathermap.org/data/2.5/weather';\n\nawait axios.get(url, {\n params: {\n q: input,\n units: 'metric',\n appid: api_key,\n },\n});
```

q is the city name. units=metric requests Celsius temperature and metric measurements. appid supplies the API key. Axios performs the HTTP GET request and returns a response object.

Example request structure (placeholder only)

```
https://api.openweathermap.org/data/2.5/weather?q=Ahmedabad&units=metric&appid=YOUR_API_KEY
```

D. Success and error handling

```
.then((res) => {\n setWeather({\n data: res.data,\n loading: false,\n error: false,\n });\n})
```

res.data is the response body. Saving it in state causes React to render the new weather. A catch handler handles failures. A more robust app should distinguish invalid cities from network errors, invalid API keys and rate limits.

4. Understanding the API fields

Expression	Meaning
weather.data.name	City name returned by the API
weather.data.sys.country	Country code
weather.data.main.temp	Temperature in requested units
weather.data.weather[0].description	Weather description, e.g. clear sky
weather.data.weather[0].icon	Code used to construct the weather icon URL
weather.data.wind.speed	Wind speed, normally metres per second for metric units

The temperature may be rounded with Math.round(). The displayed date is generated by JavaScript using the computer's local date/time, while weather data comes from the API. Small differences from a phone app can occur because of location precision, update timing, provider or rounding.

5. Project file map

src/App.js — main UI logic, state and API request.
src/App.css — layout, colors, spacing and styling.
src/index.js — React entry point; mounts the app into the HTML page.
package.json — dependencies and scripts.
public/ — HTML template and static assets.
src/Node/index.js — separate Node.js server example; not needed for the weather app.

6. How to run the project again

```
node -v\nnpm -v\ncd path\\to\\ETT\nnpm install\nnpm start
```

Run commands in the folder containing package.json. The app normally opens at http://localhost:3000. Keep the terminal running. Press Ctrl+C to stop the server. Restart the server after changing a .env value.

7. Common viva questions

Q1. What is React?
A JavaScript library for building user interfaces from reusable components.

Q2. What is Axios?
A JavaScript HTTP client used to request weather data from an API.

Q3. What is an API?
A defined way for one software system to request data or services from another. OpenWeather supplies the weather information.

Q4. Why useState?
It stores changing values and tells React to update the interface when they change.

Q5. What is JSON?
A text format for structured data, such as the weather response.

Q6. Why units=metric?
It requests Celsius temperature and metric measurements.

Q7. Is weather stored in a database in this project?
No. The app requests current data from OpenWeather and displays it; it does not need its own database for this feature.

8. Quick revision checklist

■ I can run npm start.
■ I can explain useState and the input handler.
■ I can explain the Axios GET request and its parameters.
■ I know where temperature, description and wind speed are in the JSON response.
■ I know how success and errors update the UI.
■ I know not to commit .env or expose API keys.

Suggested 30-second project explanation

“My Weather App is built with React.js. The user enters a city name, and the application uses Axios to send a GET request to the OpenWeather API. The API returns weather data in JSON format. I store the response in React state and display the city, temperature, weather condition, icon and wind speed. The interface also handles loading and request errors.”